SES598 Homework 2: Cart Pole Optimal Control

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1 Background

This assignment tunes and analyzes an LQR controller for a cart-pole system subject to earthquake disturbances. The objective of the assignment is to maintain the pole's stability while keeping the cart within its physical constraints under external perturbation from the earthquake force generator [1].

Initial simulation constraints:

Table 1: Simulation Parameters

Parameter	**Value**
Cart Traversal Range	+/-2.5 m
Cart Mass	1 kg
Pole Length	1 kg
Pole Mass	1 kg

The provided system includes an earthquake force generator that introduces external disturbances [1]:

- Generates continuous, earthquake-like forces using superposition of sine waves
- Base amplitude: 15.0N (default setting)
- Frequency range: 0.5-4.0 Hz (default setting)
- Random variations in amplitude and phase
- Additional Gaussian noise

1.1 LQR Formulation

Per lecture notes [2] provided in SES598, the LQR equation is formulated as follows.

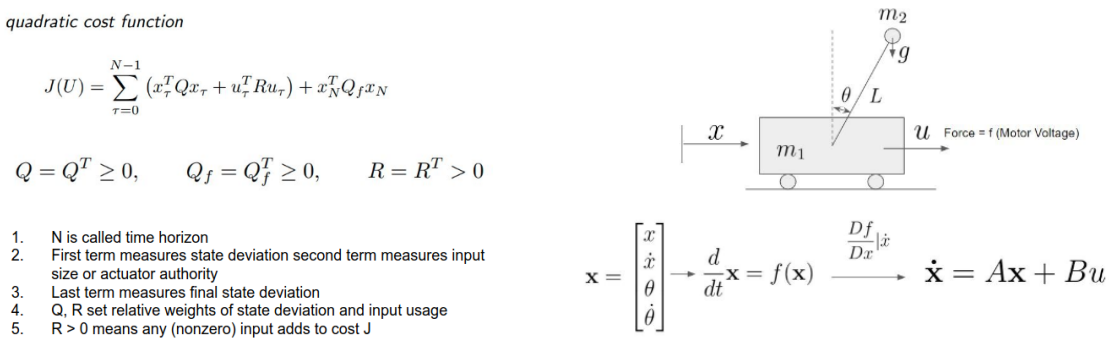

Figure 1: LQR Formulation [2]

The variable x can be written in state space form, where the cart displacement, cart velocity, pole angle, and pole angular velocity are accounted for. We want to provide a value of u, a control input, that minimizes this cost function.

The Q matrix is a gain matrix, and allows us to weight the impact of the state space variables. Increasing a term penalizes it.

1.2 Initial Controller Performance

Below are the plots associated with the original controller. This controller does not keep the cart within 2.5 m of the center and fails almost immediately.

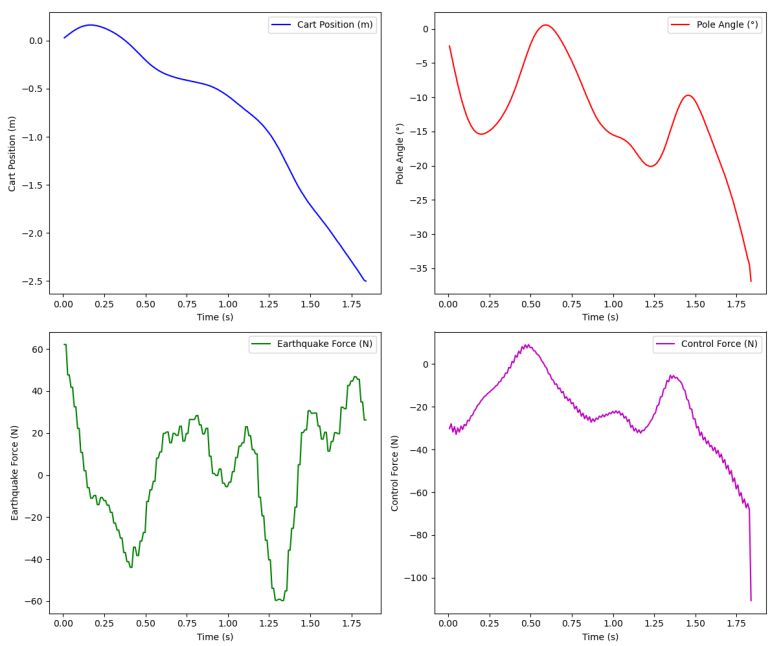

Figure 2: Default Controller with Q:[1,1,1,1] and R:[1]

2 Tuning Methodology

The controller was tuned manually through trial and error to understand the impact of different values within the Q and R matrices. The values initially started as small single entries within Q and were gradually increased to understand the impact of the parameter on performance.

It is not efficient to constantly adjust each parameter each time. It is beneficial to start with a single matrix, and then move to the subsequent matrix once it is operating proficiently and repeat the process.

The order of recommended parameter tuning is to start with the Q matrix then move to manually tuning the R matrix. Increasing several parameters within this matrix proved to be most successful.

2.1 Different Parameter Comparison

2.1.1 Changing Values of Q

To understand the impact of the values of Q, each of the entries were adjusted. Increasing only one of the entries does not produce a significant improvement in control so increasing combinations of different parameters is utilized. The value of R is constant for all the trials in Table 2, R is a value of 1.0.

Table 2: Performance of Different Q Matrices

Q Values	**Duration of Stability (s)**	**Max Cart Displacement (m)**	**Max Angle Deviation (Deg)**	**Avg. Control Effort (N)**	**Stability Score**
[10, 1, 1, 1]	3.34	2.5	9.235	11.48	2.58
[10, 10, 1, 1]	78.65	2.5	6.009	1.362	3.73
[10, 1, 10, 1]	5.59	2.5	12.896	11.537	1.84
[10, 1, 1, 10]	12.33	2.5	6.568	4.526	3.46
[10, 20, 1, 1]	120.01	0.410	1.387	0.457	8.88
[10, 20, 1, 5]	120	0.131	0.801	0.454	9.55

The first entry in the Q matrix weights the significance of the cart displacement from the center of the track, the second term accounts for the significance of the carts velocity and how fast it moves, the third entry pertains to the pole angle, and the last entry pertains to how fast the pole moves/swings. Increasing the value of an entry in Q further penalizes the term. A combination of increased terms proved most fruitful in improving the performance.

The Q matrix [10, 20, 1, 5] provided the best performance and stability of the cart. The Q matrix prioritizes the speed of the cart, specifically a slower speed.

2.1.2 Changing Values of R

Below are different trials of varying values of R with the Q matrix determined in the previous section, [10, 20, 1, 5].

Table 3: Performance of Different R Matrices

R Values	**Duration of Stability (s)**	**Max Cart Displacement (m)**	**Max Angle Deviation (Deg)**	**Avg. Control Effort (N)**	**Stability Score**
0.1	120.00	0.022	0.340	0.341	9.87
0.25	120.00	0.061	1.682	2.264	9.43
0.5	120.01	0.066	1.252	1.534	9.54
3	9.41	2.5	10.406	9.683	2.43

The value of R seems to produce better stability overall when it is smaller. In general, as the value of R increases, the average control effort also increases. Keeping the value of R seems to allow the controller to use force more freely, and apply control sooner.

3 Optimized Controller

The controller with the previously determined values of Q and R is seen below.

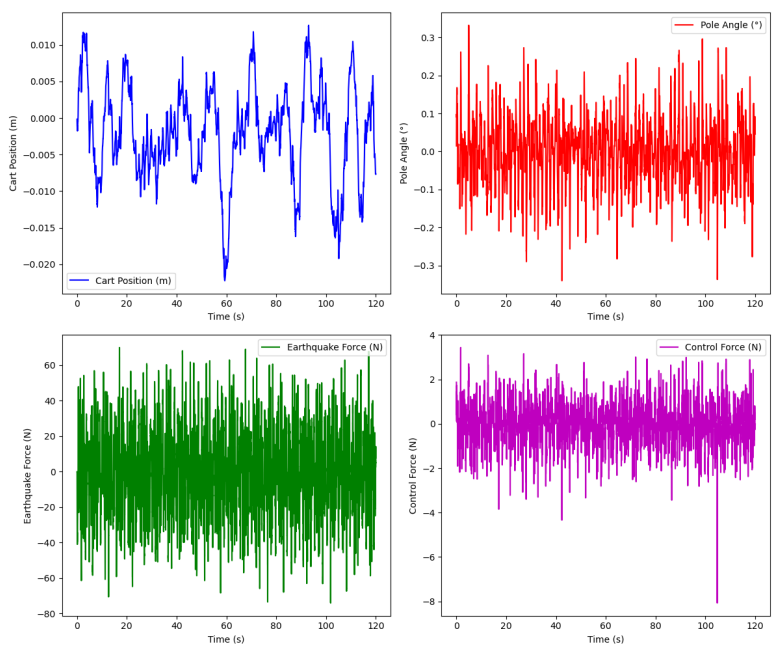

Figure 3: Optimized Controller with Q:[10,20,1,5] and R:[0.1]

As shown above, the cart stays very close to the center of the track and the pole moves very little in angle as well. There are a lot of small, control forces applied to keep the cart stable over time.

4 Conclusion

In conclusion, the cart pole problem successfully achieved stability for the 120 second simulation. The simulation of this video can be found in the repository, linked under reference [3] below.

5 References

[1] Course Repo: https://github.com/gciaglia/ses598-space-robotics-and-ai-2026/blob/main/assignments/cart_pole_optimal_control

[2] Das, J. (2026, January 22–27). State space models, state estimation, Kalman filtering (Lecture 4–5) [Course lecture notes]. SES 598: Space Robotics and AI, [University Name]. https://canvas.asu.edu/courses/242061/files/folder/Lecture%20slides?preview=125431657

[3]Video Submission: https://github.com/gciaglia/ses598-space-robotics-and-ai-2026/blob/main/assignments/cart_pole_optimal_control/Final%20Submission/OptimizedLQR_VideoSubmission.mp4